

Curriculum Vitae

Saroj Chand, Postdoctoral Scholar

+1 (347) 459-4413 • schand@berkeley.edu • [Google Scholar](https://scholar.google.com/citations?user=sarojc23) • sarojc23.github.io

Research Interests

Quantum information science and engineering; solid-state spin qubits and quantum sensing (NV centers, hBN VB⁻ and related defects); coherent spin control, quantum noise spectroscopy, and optical/ODMR spectroscopy; 2D quantum materials and correlated quantum matter (TMDs, hBN with color centers, magnetic vdW systems); quantum photonics/nanophotonics for spin–photon interfaces and quantum instrumentation; microwave/RF engineering and cryogenic platforms for quantum control; high-performance computing and first-principles (DFT) modeling for quantum materials and device design.

Education and Training

- **Postdoctoral Scholar**, University of California, Berkeley, Nov 2023–*Present*
Advisor: **Prof. Shimon Kolkowitz**
- **Ph.D. in Physics**, The Graduate Center, City University of New York (CUNY), Oct 2023
Dissertation: *Exciton Dynamics, Interaction, and Transport in Monolayers of Transition-Metal Dichalcogenides*
Advisor: **Prof. Gabriele Grosso**
- **M.Phil. in Physics**, The Graduate Center, CUNY, Nov 2022
- **M.Sc. in Physics**, Tribhuvan University, Kathmandu, Nepal, 2016
- **B.Sc. in Physics**, Tribhuvan University, Kathmandu, Nepal, 2013

Research Experience

Postdoctoral Scholar, University of California, Berkeley (Berkeley, CA)

Nov 2023 – Present, Faculty Advisor: Prof. Shimon Kolkowitz

- **Built** a multiplexed wide-field NV microscope **with 100+ individually resolved single-defect control and readout for scalable magnetometry (featured in Phys. Rev. X 15, 031015 (2025))**.
- Deployed **ODMR, dynamical decoupling, state tomography** in high-throughput workflows → higher sensitivity & reproducibility.
- Engineered **optical (SLM/AOD/EMCCD, cryo)** and **microwave** chains; added **automation/feedback** to stabilize alignment & boost throughput.
- **Mentored** undergrad/grad researchers; coordinated photonics–electronics–cryo builds to accelerate **build–measure–analyze** cycles.

Graduate Research Assistant, Advanced Science Research Center (CUNY) (New York, NY)

Aug 2018 – Oct 2023, Faculty Advisor: Prof. Gabriele Grosso

- Studied exciton dynamics and quantum emitters in TMDs and hBN; integrated van der Waals heterostructures into photonic structures and identified emergent excitonic states in **phase-engineered 1H/1T monolayer**.
- Led the development of a **cryogenic confocal single-photon microscope** (APDs/HBT + spectrometers) for low-temperature PL/absorption (4–300 K); **expanded** to a **dual-galvo** scanner and a **wide-field** imaging path.

- Led/ contributed collaborations (Photonics Initiative, national labs, Flatiron Institute, engineering department) on metamaterials, plasmonics, photonic structures, and DBRs, enabling strong light–matter coupling → **4 publications + 4 manuscripts under review/ prep.**
- Designed and fabricated plasmonic/dielectric nanostructures (e-beam lithography, etch, atomic-layer deposition) and integrated automated control for high-throughput optical measurements.

Guest Researcher (External Collaborator), Flatiron Institute – Simons Foundation (New York, NY)

Aug 2019 – Dec 2023, Collaborator: Prof. Johannes Flick

- Performed **DFT/DFPT & excited-state** calculations (VASP, QE, Wannier90) on HPC for 2D & wide-bandgap materials.
- Modeled **excitons & electron–phonon coupling**; interpreted PL/absorption and defect signatures.
- Built scalable **HPC workflows** (job arrays, auto-convergence, post-processing) for large-supercell/defect studies.

Publications

Peer-reviewed

1. M. Cambria*, **S. Chand*** (co-first authors), C. Reiter, S. Kolkowitz, “Scalable Parallel Measurement of Individual Nitrogen-Vacancy Centers,” **Phys. Rev. X** 15, 031015 (2025). *Featured in Physics*
2. **S. B. Chand**[†], J. M. Woods, J. Quan, E. Mejia, T. Taniguchi, K. Watanabe, A. Alù, G. Grosso[†], “Phonon-Dressed States of Dark Excitons in Transition Metal Dichalcogenides” *Advanced Optical Material* (2026). [†]*Corresponding Authors*
3. **S. B. Chand**, J. M. Woods, J. Quan, E. Mejia, T. Taniguchi, K. Watanabe, A. Alù, G. Grosso, “Interaction-driven transport of dark excitons in 2D semiconductors with phonon-mediated optical readout,” **Nat. Commun.** 14, 3712 (2023).
4. **S. B. Chand**, J. M. Woods, E. Mejia, T. Taniguchi, K. Watanabe, G. Grosso, “Visualization of dark excitons in semiconductor monolayers for high-sensitivity strain sensing,” **Nano Letters** 22(7), 3087–3094 (2022).
5. J. M. Woods*, **S. B. Chand*** (co-first authors), E. Mejia, T. Taniguchi, K. Watanabe, J. Flick, G. Grosso, “Emergence of a low-energy excitonic state in single-layer WS₂ with 1H/1T phase mixture,” **ACS Photonics** 11(9), 3784–3793 (2024).
6. J. Quan, M. Cotrufo, **S. B. Chand**, Z. Liu, D. Huang, T. Taniguchi, K. Watanabe, G. Grosso, X. Li, A. Alù, “*Plasmonic Heterostructure for On-Site Enhancement and Control of Spin-Forbidden Dark Excitons in Monolayer Semiconductors*,” **Nature Photonics** (2025).
7. S. Guddala, S. Kiriushchikina, Y. Kawaguchi, A. Vakulenko, **S. B. Chand**, E. M. Hobuss, G. Grosso, A. Alù, A. B. Khanikaev, “*Topological Polaritonic Metamaterials Realized in Bulk Transition Metal Dichalcogenide Crystals*,” **Small Structures** (2025).
8. M. Cho, B. Datta, K. Han, **S. B. Chand**, P. C. Adak, S. Yu, F. Li, K. Watanabe, T. Taniguchi, J. Hone, J. Jung, G. Grosso, Y. D. Kim, V. M. Menon, “Moiré exciton polaron engineering via twisted hBN,” **Nano Letters** 24, XX (2024).
9. J. Pelliciari, E. Mejia, J. M. Woods, Y. Gu, J. Li, **S. B. Chand**, S. Fan, K. Watanabe, T. Taniguchi, V. Bisogni, G. Grosso, “Elementary excitations of quantum emitters in hexagonal boron nitride,” **Nature Materials** 23, 1230–1236 (2024)
10. E. A. Mejia, J. M. Woods, **S. B. Chand**, E. Ramjattan, T. Taniguchi, K. Watanabe, J. Pelliciari, G. Grosso, “General algorithm for characterization of donor–acceptor pair recombination processes in solid-state materials,” **Opt. Materials Express** 14(9), 2122–2133 (2024).

11. M. Wang, G. Hu, **S. B. Chand**, M. Cotrufo, Y. Abate, K. Watanabe, T. Taniguchi, G. Grosso, C.W. Qiu, A. Alù, “Spin–orbit-locked hyperbolic polariton vortices carrying reconfigurable topological charges,” **eLight** 2, 12 (2022).

Under review / in preparation

12. **S. B. Chand**, S. Kolkowitz, “*Multiplexed tomography of the ^{13}C spin environment of hundreds of individual NV centers*” – *In preparation* (2025).
13. M. Wang, S. Yin, **S. B. Chand**, J. Quan, X. Jiang, K. Watanabe, T. Taniguchi, G. Grosso, A. Alù, “*Mid-Infrared Long-Range Surface Polaritons in Two-Dimensional van der Waals Materials*,” – *Under Review* (2025).
14. X. Jiang, J. Quan, S. Yin, **S. Chand**, S. Guo, Z. Liu, M. Cotrufo, G. Grosso, X. Li, A. Alù, “*Deep-Subwavelength Bucket Nanocavity for Ultralow-Threshold Interlayer Exciton Lasing*” – *In Preparation* (2025).

Presentations

Invited Talks & Research Seminars

1. “**What Can 1,000 Quantum Sensors See That One Cannot?: Building Scalable Quantum Hardware with Individually Resolved NV Centers.**” Quantum Gathering, UC Berkeley, May 2026.
2. “**Multiplexed Quantum Sensing with Solid-State Spins and Quantum Materials.**” Florida International University, Colloquium, April 2026.
3. “**Multiplexed Quantum Sensing with Solid-State Spins and Quantum Materials.**” Louisiana State University, Colloquium, March 2026.
4. “**Multiplexed Quantum Sensing with Solid-State Spins and Quantum Materials.**” University at Buffalo, SUNY, Colloquium, February 2026.
5. “**Multiplexed NV Centers and Solid-State Emitters in 2D Materials.**” Center for Nanoscale Materials, Argonne National Laboratory, NST Division, 2025. Invited in-person seminar.
6. “**Exciton Dynamics and Transport in TMD Monolayers.**” Pritzker School of Molecular Engineering, University of Chicago, 2023. Invited special seminar.
7. “**Exciton Dynamics and Transport in 2D Semiconductors.**” Virtual research seminars, 2023:
 - California Institute of Technology, Hadt Lab
 - University of Washington, Kai-Mei Fu Group
8. “ **$1\text{H}/1\text{T}$ Phase Engineering in WS_2 : Pathways to Novel Optical Response.**” CUNY ASRC Science Café, 2021. Public outreach talk.

Contributed Conference Talks

5. “**Parallel Control and Measurement of Over 100 Nitrogen-Vacancy Centers in Diamond.**” APS DAMOP Meeting, 2025.
6. “**Visualizing Momentum-Forbidden Dark Excitons in Layered Semiconductors.**” CLEO: Conference on Lasers and Electro-Optics, 2023.
7. “**Interaction-Driven Transport of Dark Excitons in 2D Semiconductors.**” APS March Meeting, 2023. *(oral)*
8. “**Visualizing Momentum-Forbidden Dark Excitons for Strain Sensing.**” ICSCE-11: International Conference on Spontaneous Coherence of Excitons, 2022.

Posters

1. “**Scalable Parallel Measurement and Control of Individual Nitrogen-Vacancy Centers.**” CIQC Annual Retreat, **2025**.
2. “**Scalable Parallel Measurement and Control of Individual Nitrogen-Vacancy Centers.**” UC Berkeley Graduate Research Fair, **2024**.
3. “**Superlattices via Defect Engineering in Two-Dimensional Semiconductors.**” *Graphene for US* Workshop, **2019**.

Teaching and Mentoring Experience

- **UC Berkeley (2023–present):** Mentor to **2 Ph.D.** and **4 B.S.** students (quantum sensing & NV control).
- **CUNY ASRC (2019 Led the development2023):** Mentor to **1 Ph.D.** and **2 B.S.** students
- **Adjunct Lecturer**, City College of New York (CUNY), 08/2019–10/2023
 - Taught undergraduate lab courses: **Optics, Electronics, Classical Mechanics, Electromagnetism** (typical section size ~20–30; online cohort up to ~66).
 - Helped to modernize labs with simulation & analysis notebooks; coordinated multi-section grading & TA teams.

Honors and Awards

- Photonics Initiative **Tithe Fellowship** (Inaugural Recipient), CUNY Advanced Science Research Center – 2023
- **Early Research Initiative (ERI) Pre-Dissertation Science Award**, CUNY Graduate Center – 2021
- “**Converge to Transform**” **Twitter Poster Session Winner**, ASRC CUNY – 2020
- **CUNY Science Scholarship**, CUNY Graduate Center – 2018

Professional Service & Outreach

- **Peer Review:** *Physical Review Letter*, *Applied Physics Letter*, *Phys. Rev. Applied*, *ACS Omega*, *IEEE Sensors* and related journals (2022–present).
- **Session Chair**, *APS DAMOP 2025*, *Portland, OR, USA*, “Quantum Sensing & Metrology III” session.
- **Co-Organizer, Quantum Gatherings** (NSF CIQC): biweekly seminar & journal club; invite speakers, curate topics, lead discussions, cross-department outreach.
- **Doctoral Students’ Council Representative (Physics)**, CUNY Graduate Center (2019–2023): organized student seminars; advocacy & community building.
- Outreach: science cafés, public talks, and mentoring of first-gen & international students.

References

Shimon Kolkowitz (Postdoctoral Advisor)

Associate Professor and Herst Chair in Physics, Department of Physics, University of California, Berkeley
Email: kolkowitz@berkeley.edu · Tel: +1 (608) 262-2865

Gabriele Grosso (Ph.D. thesis advisor)

Associate Professor of Physics, Photonics Initiative, CUNY Advanced Science Research Center; Graduate Center of CUNY
Email: ggrosso@gc.cuny.edu · Tel: +1 (212) 413-3272

Andrea Alù (Collaborator/mentor)

Founding Director, Photonics Initiative, CUNY Advanced Science Research Center;

Einstein Professor of Physics, Graduate Center of CUNY; Professor of Electrical Engineering, City College of New York

Email: aalu@gc.cuny.edu · Tel: +1 (212) 413-3260

Dmitry Budker (Collaborator/mentor)

Professor Emeritus of the Graduate School, Department of Physics, University of California, Berkeley

Email: budker@berkeley.edu · Tel: +1 (608) 262-2865